

Practice Questions for Complementary Variables

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1 Questions

1. In the context of complementary slackness, what is the relationship between the primal and dual variables if a constraint in the primal problem is binding (i.e., the inequality holds with equality)?
 - a. The corresponding dual variable is zero.
 - b. The corresponding dual variable is positive.
 - c. The corresponding primal variable is zero.
 - d. There is no specific relationship between the primal and dual variables.
 - e. Both the corresponding primal and dual variables are zero.
2. Explain the concept of complementary slackness and how it helps in verifying the optimality of solutions in linear programming.
 - a. Complementary slackness states that the optimal objective function values of the primal and dual problems are equal.
 - b. Complementary slackness states that at optimality, either a primal variable or its corresponding dual slack variable must be zero, and similarly for dual variables and primal slack variables.
 - c. Complementary slackness is a method for finding an initial feasible solution for the simplex method.
 - d. Complementary slackness is a pivot rule used in the simplex method to select the entering variable.
 - e. Complementary slackness is a condition that must be met for a linear programming problem to be feasible.
3. Suppose you have solved a primal linear programming problem and its dual and found optimal solutions. How can complementary slackness be used to verify the optimality of the solutions?
 - a. Check if the optimal objective function values of the primal and dual problems are equal.
 - b. Check if all constraints in both the primal and dual problems are binding.
 - c. Check if, for each constraint in the primal problem, either the constraint is binding or the corresponding dual variable is zero, and vice versa for the dual problem.
 - d. Check if all variables in both the primal and dual problems are non-negative.
 - e. Check if the sum of the primal and dual objective function values is zero.

4. What is the practical implication of complementary slackness in sensitivity analysis of linear programming problems?
 - a. It helps determine the optimal solution to the dual problem.
 - b. It helps determine the range of changes in the objective function coefficients that maintain the current optimal basis.
 - c. It helps determine the range of changes in the right-hand side values that maintain the current optimal basis.
 - d. It helps identify which constraints are binding at the optimal solution and how changes to those constraints affect the optimal solution.
 - e. It helps determine the optimal solution to the primal problem.
5. Explain the concept of complementary slackness in the context of linear programming and its significance in verifying the optimality of solutions to primal and dual problems. Illustrate your explanation with a simple example.
 - a. Complementary slackness states that if a primal variable is positive, its corresponding dual slack variable must be zero, and vice versa. This condition holds for both primal and dual problems simultaneously. It helps verify optimality because if both primal and dual solutions satisfy complementary slackness, then both are optimal.
 - b. Complementary slackness is a theorem that proves the existence of an optimal solution for both the primal and dual problems. It states that if one problem has a feasible solution, the other must also have a feasible solution. It doesn't directly help in verifying optimality.
 - c. Complementary slackness is a condition that must be met only for the primal problem to be optimal. It states that if a constraint is binding (holds with equality), then the corresponding dual variable must be zero. It doesn't consider the dual problem's optimality.
 - d. Complementary slackness is a method for solving linear programming problems by iteratively improving the primal solution until it satisfies a set of conditions. It doesn't directly relate to the dual problem or optimality verification.
 - e. Complementary slackness is a concept that only applies to problems with equality constraints. It states that the optimal values of the primal and dual variables must be equal. It doesn't apply to problems with inequality constraints.
6. In the context of complementary slackness, if the i -th constraint in the primal problem is binding (i.e., the inequality holds with equality), what can we say about the corresponding dual variable y_i ?
 - a. y_i must be zero.
 - b. y_i must be positive.
 - c. y_i can be zero or positive.
 - d. y_i must be negative.
 - e. y_i can be any real number.
7. Consider the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + 3x_2 \leq 12$, $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$. Suppose an optimal solution is $x_1 = 4, x_2 = 4$ with $Z = 20$. The corresponding dual problem has an optimal solution $y_1 = 1, y_2 = 1, y_3 = 0$. Verify the optimality using complementary slackness. What is the value of $x_1 z_1$?

- a. 0
 - b. 4
 - c. 1
 - d. 20
 - e. 16
8. In the primal problem (Slide 1), what are the values of the slack variables w_1 , w_2 , and w_3 when $x_1 = 0$ and $x_2 = 0$?
- a. $w_1 = 0, w_2 = 0, w_3 = 0$
 - b. $w_1 = 12, w_2 = 8, w_3 = 10$
 - c. $w_1 = -12, w_2 = -8, w_3 = -10$
 - d. $w_1 = 8, w_2 = 12, w_3 = 10$
 - e. $w_1 = 10, w_2 = 12, w_3 = 8$
9. Using Slide 1, formulate the dual problem for the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + 3x_2 \leq 12$, $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$. What is the objective function of the dual problem?
- a. Minimize $W = 12y_1 + 8y_2 + 10y_3$
 - b. Maximize $W = 12y_1 + 8y_2 + 10y_3$
 - c. Minimize $W = 3y_1 + 2y_2$
 - d. Maximize $W = 3y_1 + 2y_2$
 - e. Minimize $W = 12y_1 + 3y_2 + 2y_3$
10. In the context of complementary slackness, if the i -th constraint in the primal problem is binding (i.e., the inequality holds with equality), what can we say about the corresponding dual variable y_i ?
- a. The corresponding dual variable y_i must be zero.
 - b. The corresponding dual variable y_i must be positive.
 - c. The corresponding dual variable y_i can be either zero or positive.
 - d. The corresponding dual variable y_i must be negative.
 - e. There is no relationship between the primal constraint and the dual variable.

2 Answers

1. In the context of complementary slackness, what is the relationship between the primal and dual variables if a constraint in the primal problem is binding (i.e., the inequality holds with equality)?
 - a. **Incorrect. A zero dual variable implies the constraint is non-binding.**
 - b. If a primal constraint is binding, it means the inequality holds with equality. Complementary slackness states that if a primal constraint is binding, then the corresponding dual variable must be positive. This is because the dual variable represents the shadow price or marginal value of relaxing that constraint. If the constraint is already at its limit (binding), then relaxing it has value, hence a positive dual variable.
 - c. **Incorrect. Complementary slackness relates primal constraints to dual variables, not primal variables.**
 - d. **Incorrect. Complementary slackness defines a specific relationship between primal constraints and dual variables.**
 - e. **Incorrect. While it's possible for both to be zero in degenerate cases, it's not a general rule for binding constraints.**
2. Explain the concept of complementary slackness and how it helps in verifying the optimality of solutions in linear programming.
 - a. **This describes strong duality, not complementary slackness.**
 - b. Complementary slackness provides a necessary and sufficient condition for optimality in linear programming. It states that at the optimal solution, for each pair of a primal variable and its corresponding dual slack variable (or a dual variable and its corresponding primal slack variable), at least one of the two must be zero. This condition helps verify the optimality of a proposed solution by checking if this condition holds for all primal and dual variables and their corresponding slack variables. If it does, then the solution is optimal.
 - c. **This describes a different technique used in linear programming.**
 - d. **This describes a different aspect of the simplex method.**
 - e. **Feasibility is determined by whether a solution satisfies all constraints, not complementary slackness.**
3. Suppose you have solved a primal linear programming problem and its dual and found optimal solutions. How can complementary slackness be used to verify the optimality of the solutions?
 - a. **This checks for strong duality, which is a necessary but not sufficient condition for verifying optimality using complementary slackness.**
 - b. **Not all constraints need to be binding at optimality.**
 - c. Complementary slackness states that for each constraint in the primal problem, either the constraint is binding (holds with equality) or the corresponding dual variable is zero. The same holds true for the dual problem: for each constraint, either it is binding or the corresponding primal variable is zero. Checking this condition for all constraints in both problems verifies the optimality of the solutions.
 - d. **Non-negativity is a condition for feasibility, not optimality via complementary slackness.**
 - e. **There's no such condition in complementary slackness.**

4. What is the practical implication of complementary slackness in sensitivity analysis of linear programming problems?
 - a. While complementary slackness is related to the dual problem, it doesn't directly determine its optimal solution.
 - b. This is related to sensitivity analysis but not directly to complementary slackness.
 - c. This is related to sensitivity analysis but not directly to complementary slackness.
 - d. In sensitivity analysis, we examine how changes in the problem's parameters affect the optimal solution. Complementary slackness helps identify which constraints are binding at the optimal solution. If a constraint is non-binding, small changes to it won't affect the optimal solution. However, changes to binding constraints will directly impact the optimal solution. Therefore, complementary slackness helps understand the sensitivity of the optimal solution to changes in the constraints.
 - e. Complementary slackness doesn't directly determine the primal optimal solution; it verifies its optimality given a primal and dual solution.

5. Explain the concept of complementary slackness in the context of linear programming and its significance in verifying the optimality of solutions to primal and dual problems. Illustrate your explanation with a simple example.
 - a. Complementary slackness is a crucial concept in linear programming duality. It states that for an optimal primal solution and its corresponding dual solution, if a primal variable is strictly positive, then the corresponding dual constraint must be binding (i.e., the slack variable is zero), and vice versa. Similarly, if a dual variable is strictly positive, then the corresponding primal constraint must be binding, and vice versa. This condition provides a powerful way to verify the optimality of solutions. If a primal and dual feasible solution satisfy complementary slackness, then both are optimal. For example, consider the primal problem: $\max z = x_1 + 2x_2$ subject to $x_1 + x_2 \leq 4$, $x_1 \leq 3$, $x_2 \leq 2$, $x_1, x_2 \geq 0$. Suppose we find a primal solution $x_1 = 2, x_2 = 2$ with $z = 6$. The corresponding dual problem is: $\min w = 4y_1 + 3y_2 + 2y_3$ subject to $y_1 + y_2 \geq 1$, $y_1 + y_3 \geq 2$, $y_1, y_2, y_3 \geq 0$. If we find a dual solution $y_1 = 1, y_2 = 0, y_3 = 1$ with $w = 6$, then complementary slackness is satisfied because $x_1 > 0$ and the first dual constraint is binding ($y_1 + y_2 = 1$), $x_2 > 0$ and the third dual constraint is binding ($y_1 + y_3 = 2$), and $y_1 > 0$ and the first primal constraint is binding ($x_1 + x_2 = 4$). Since both solutions are feasible and satisfy complementary slackness, they are optimal.
 - b. This option incorrectly states that complementary slackness proves the existence of optimal solutions. While it's a consequence of strong duality, it's not its primary purpose. Complementary slackness is a condition that helps verify the optimality of *already found* solutions.
 - c. This option incorrectly limits complementary slackness to only the primal problem. It's a condition that must hold for *both* primal and dual solutions simultaneously to guarantee optimality.
 - d. This option mischaracterizes complementary slackness as a solution method. It's not an algorithm but a condition used to verify optimality after a solution has been found.
 - e. This option incorrectly restricts complementary slackness to problems with only equality constraints. It applies equally to problems with inequality constraints, as illustrated in the correct answer.

6. In the context of complementary slackness, if the i -th constraint in the primal problem is binding (i.e., the inequality holds with equality), what can we say about the corresponding dual variable y_i ?

- a. This is incorrect. Complementary slackness states that if a primal constraint is binding, the corresponding dual variable is positive. If the primal constraint is not binding, then the dual variable is zero.
 - b. If the i -th primal constraint is binding, then the corresponding dual variable y_i must be positive. This is a direct consequence of complementary slackness.
 - c. This is incorrect. Complementary slackness dictates that only one of the pair (primal slack, dual variable) can be non-zero. Since the primal constraint is binding, the slack is zero, thus the dual variable must be positive.
 - d. Dual variables are non-negative in the standard form of the dual problem. A negative dual variable would violate the dual problem's constraints.
 - e. Dual variables are non-negative. Complementary slackness further restricts their values based on the primal solution.
7. Consider the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + 3x_2 \leq 12$, $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$. Suppose an optimal solution is $x_1 = 4, x_2 = 4$ with $Z = 20$. The corresponding dual problem has an optimal solution $y_1 = 1, y_2 = 1, y_3 = 0$. Verify the optimality using complementary slackness. What is the value of $x_1 z_1$?
- a. Since $x_1 = 4$ and the first constraint is binding ($-x_1 + 3x_2 = 8$), the corresponding dual variable $y_1 = 1$. The dual slack variable z_1 must be zero due to complementary slackness. Therefore, $x_1 z_1 = 4 \times 0 = 0$.
 - b. This is incorrect. It uses the value of x_1 without considering the complementary slackness condition.
 - c. This is incorrect. It uses the value of y_1 without considering the complementary slackness condition.
 - d. This is incorrect. It uses the value of the objective function Z .
 - e. This is incorrect. It uses the product of x_1 and x_2 without considering the complementary slackness condition.
8. In the primal problem (Slide 1), what are the values of the slack variables w_1 , w_2 , and w_3 when $x_1 = 0$ and $x_2 = 0$?
- a. This is incorrect. The slack variables represent the difference between the left and right sides of the constraints. Since the left side is 0, the slack variables equal the right-hand side values.
 - b. When $x_1 = 0$ and $x_2 = 0$, the constraints become $w_1 = 12$, $w_2 = 8$, and $w_3 = 10$. These are the values of the slack variables at the origin.
 - c. This is incorrect. Slack variables are always non-negative in the standard form of a linear program.
 - d. This is incorrect. The values of the slack variables are directly obtained by substituting $x_1 = 0$ and $x_2 = 0$ into the constraints.
 - e. This is incorrect. The values of the slack variables are directly obtained by substituting $x_1 = 0$ and $x_2 = 0$ into the constraints.
9. Using Slide 1, formulate the dual problem for the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + 3x_2 \leq 12$, $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$. What is the objective function of the dual problem?

- a. The dual problem is a minimization problem. The objective function coefficients in the dual problem are the right-hand side constants from the primal problem's constraints. Therefore, the objective function is Minimize $W = 12y_1 + 8y_2 + 10y_3$.
 - b. The dual of a maximization problem is a minimization problem.
 - c. These are the coefficients from the primal objective function, not the dual.
 - d. The dual of a maximization problem is a minimization problem.
 - e. The coefficients are incorrect; they should be the right-hand side values from the primal constraints.
10. In the context of complementary slackness, if the i -th constraint in the primal problem is binding (i.e., the inequality holds with equality), what can we say about the corresponding dual variable y_i ?
- a. Incorrect. While it's possible for y_i to be zero, it's not mandatory. Complementary slackness allows for y_i to be positive when the constraint is binding.
 - b. Incorrect. While y_i can be positive, it's not always the case. Complementary slackness allows for y_i to be zero when the constraint is binding.
 - c. The corresponding dual variable y_i can be either zero or positive. Complementary slackness states that either the primal constraint is binding ($a_i^T x = b_i$) or the corresponding dual variable is zero ($y_i = 0$), but not both. If the constraint is binding, y_i can be zero or positive.
 - d. Incorrect. Dual variables associated with less-than-or-equal-to constraints in a maximization problem are non-negative. Therefore, y_i cannot be negative.
 - e. Incorrect. Complementary slackness establishes a direct relationship between primal constraints and dual variables.